Operations & User Manual (OUM)

SQRT (Sub-zero Quantitative Research Team)

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Document History

Date	Version	Notes	Approver
2026–04–15	SQT-FAR-OUM-0	Initial release	EA

Author Contributions

OND §1–7

JA §3,7

CK §2,3,7

BR §1,3,7

EA §5, Reviewer

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Key Reference Documents

1. UCD-DRD-2026-0
2. SQT-CDR-SEP-0
3. SQT-CDR-PMP-0

1 Executive Summary

This Operations & User Manual (OUM) document provides a clear and structured operational guide for the SQRT mission payload from final assembly through to launch and post-flight recovery.

Section 2 of this document includes an overview of the SQRT mission, including goals and requirements. Section 3 provides an overview of the full payload system, including both hardware and software components. Operator health and safety information is included in Section 4, and satellite safety and handling information is included in Section 5. An overview of operations procedures is included in Section 6, with further details in the document's Appendix.

The Appendix includes a full prior assembly and handling advice, a complete tool and equipment list required on launch day, launch and recovery procedures, flight operations procedures, necessary deactivation steps to be followed for the preservation of mission data post-flight. SQRT Contact personnel are also listed.

2 Mission Overview

Mission Statement:
The SQRT mission will produce and observe the flash evaporation of water via exposure to a low-pressure and low-temperature environment to investigate how such plume-like vapour structures thermally evolve.

This flash evaporation process is directly relevant to cryovolcanic plumes on icy moons in our solar system as well as de-salinisation plants on Earth. In order to reproduce the conditions required for flash evaporation, an isolated water container will be flown up to altitudes of at least 20km, where temperatures and pressures are much lower. Upon sudden exposure to this environment, the water should instantly undergo a phase change into gas from its liquid form provided that the water is in a liquid state at the time of exposure.

The mission objectives focus on observing the thermal evolution of the water sample undergoing this flash evaporation process.

2.1 Mission Goals

MO1: Fly and recover a TupperSat containing an experimental scientific or engineering/technology demonstration payload with potential application to space science.

MO2: Reproduce and observe the flash evaporation process and subsequent thermal evolution of water when exposed to near-space conditions.

2.2 Level 1: Mission Requirements

2.2.1 Customer (UCD) Requirements

Table 1: Customer (UCD) Mission Requirements

Requirement ID	Title	Description	Justification
UCD:MR:0010	Payload Size	The TupperSat shall be carried as a payload on a light unmanned free balloon.	Inherited from UCD-PRD-2026-0.
UCD:MR:0020	Flight Time	The TupperSat shall operate for at least 6 hours.	Inherited from UCD-PRD-2026-0.
UCD:MR:0030	Flight Environment	The TupperSat shall operate up to 35 km altitude.	Inherited from UCD-PRD-2026-0.
UCD:MR:0040	TDRS Communication	The TupperSat shall communicate with a ground station using TDRS.	Inherited from UCD-PRD-2026-0.
UCD:MR:0050	Tracking and Recovery	The TupperSat shall report real-time position telemetry.	Inherited from UCD-PRD-2026-0.
UCD:MR:0060	Non-Recovery	The TupperSat shall transmit any experimental data necessary to meet the mission objectives in the event that recovery is not possible.	Inherited from UCD-PRD-2026-0.
UCD:MR:0070	Atmospheric Profiling	The TupperSat shall measure and report the atmosphere's temperature and pressure profiled with vertical resolution of at least 1 measurement per 200 m ascent (goal: at least 1 measurement per 100 m).	Inherited from UCD-PRD-2026-0.

UCD:MR:0080	Launch Readiness	The TupperSat shall be delivered ready to be turned on for launch without further intervention.	Inherited from UCD-PRD-2026-0.

2.2.2 Mission Specific Requirements

Table 2: Mission Specific Requirements

Requirement ID	Title	Description	Justification
SQT:MR:010	Thermal Data	The TupperSat shall measure the temperature of the water and/or surrounding vapour during exposure to reduced pressure	To characterise and quantify its thermal evolution and identify key transition events.
SQT:MR:030	Timestamped Data	The TupperSat shall record the time at each primary measurement.	Storing timestamps at each data acquisition will facilitate chronological results.
SQT:MR:040	Altitude Data	The TupperSat shall record the altitude for all primary data acquisitions.	Recording the altitude will verify if the trigger sensor and exposure mechanism functioned at the desired altitude.
SQT:MR:050	Trigger Operation	The TupperSat shall provide verifiable confirmation that the trigger condition has been met	This requirement supports UCD:MR:0060 for the scenario of non-recovery of the TupperSat.
SQT:MR:060	Exposed Sample Containment	The TupperSat shall maintain the water sample within the low-pressure chamber for at least 2 seconds	This requirement allows sufficient time for the primary data to be acquired.
SQT:MR:070	External Environment Readings	The TupperSat shall record continuous external pressure data during the exposure period.	This requirement will aid post-mission analysis. External pressure can be compared to the pressures inside the low-pressure chamber to appraise performance of venting.
SQT:MR:080	Ease of Data Retrieval	The TupperSat shall be designed in such a way that primary experimental data is recorded, stored and retrievable following payload recovery	For post-mission analysis and subsequent confirmation of mission success. This process will be covered in the DMP.
SQT:MR:090	Viability for Prototyping	The TupperSat shall be designed to allow repeatable ground-based testing	To ensure reliability and durability of mission-critical subsystems.
SQT:MR:100	Water Exposure	The TupperSat shall be designed in such a way to expose liquid at atmospheric pressure of ≤ 35 mbar at approx. 23 km altitude.	This requirement helps to satisfy MO2 and to ensure that flash evaporation occurs.

2.3 Level 2: Technical Requirements

3.3.1.1 Structure

Table 3: Structural Requirements

Requirement ID	Title	Description	Justification
UCD:TR:00010	TupperSat Mass	The TupperSat's total mass shall not exceed 0.8 kg.	Inherited from UCD-PRD-2026-0.
UCD:TR:00020	TupperSat Volume	The TupperSat shall be enclosed in a Tupperware-like container, with volume less than 2 litres.	Inherited from UCD-PRD-2026-0.
UCD:TR:00030	Launch Interface	The TupperSat shall attach to the launch balloon using a single carabiner connector.	Inherited from UCD-PRD-2026-0.

3 System Description & Overview

3.1 Hardware

This section provides an illustrated overview of the system, including the functionality of different components of the SQRT payload. The overall system may be divided into different subsystems, all responsible for different key operations during the flight.

The primary structure consists of a Tupperware enclosure box with four eye-hooks for support and attachment to the high altitude balloon structure. All internal subsytems are attached to metal threaded rods that connect on the outsides of a plastic tupperware outer shell. The communications antenna is attached and extruded from below the payload and the external temperature sensor extrudes from the top of the lid. The interior structure is divided into two compartments separated by thermal insulation and a 3-d printed divider: the electronics bus and the vented low pressure chamber. Figure 1 shows all components used to construct the TupperSat, laid out unassembled.

When the valve is opened, water can access the low pressure chamber via a small hole in the central divider. The experimental chamber also has a carefully corner positioned MLX90640 thermal sensor used to monitor thermal evolution of water in the chamber. This sensor is mounted in a custom printed shell which is fixed to the TupperSat structure via screws attached to a plastic arm coming from the electronics bus section of the payload, as shown in Figure 4.

The electronics bus subsystem is layered in the following order from top to bottom: On top is the cover for the electronics bus which ensures protection for the electronics from the risks of potential water exposure. Below that is the electronics board which includes all of the housekeeping sensors for measurement of altitude, atmospheric pressure, internal payload temperature and system health parameters, and Raspberry Pi Pico microcontroller and the SD storage card for all onboard data storage. Below that layer is the system power/battery pack (which is mounted using a custom printed holder shown in Figure 5). Below that is the syringe/water container which is adjoined by the silicone tubing and restricted by the servo motor to control the flow of water.

The electronics bus and low pressure chamber are isolated from each other by both plastic walls and thermal insulation. This insulating material can be seen as the silver material in Figure 2. There is now additional foam insulation material around the outside of the tupperware which is not shown in that figure. The additional foam around the assembled Tuppersat can be seen in Figure 17. The two sections of the payload are isolated from each other to ensure that a necessary above-freezing temperature is maintained within the electronics bus while the low pressure chamber is as close to outside conditions as possible. Additionally, the isolating material is in place to prevent water from reaching and interfering with the electronics.

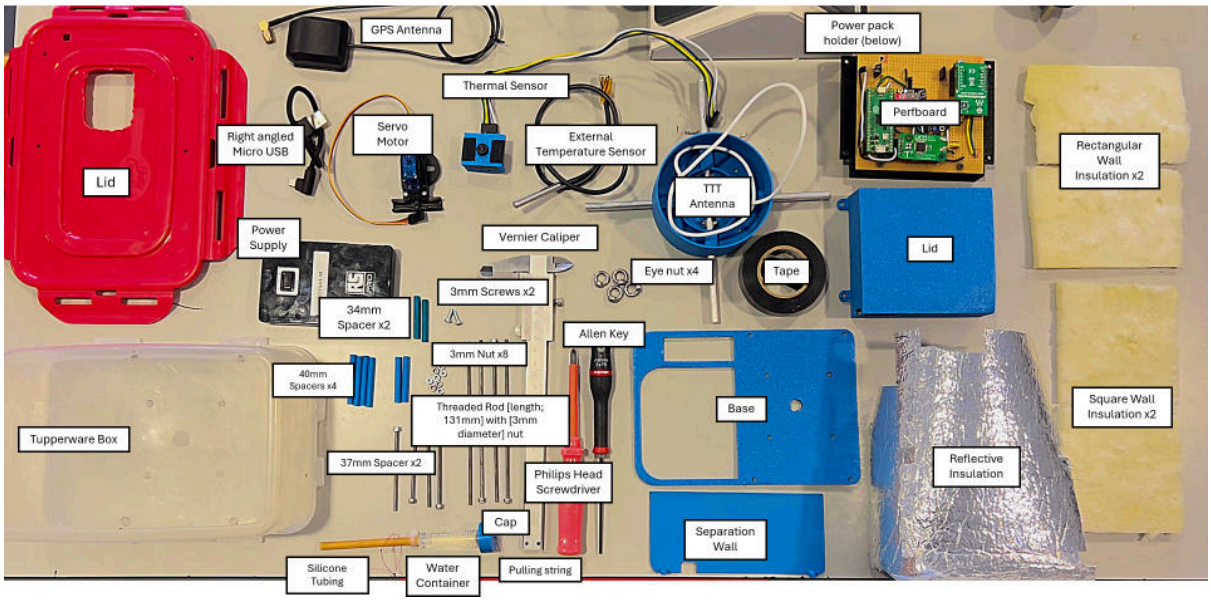


Figure 1: All labelled SQRT TupperSat components



Figure 2: Photo of Assembled SQRT Tuppersat

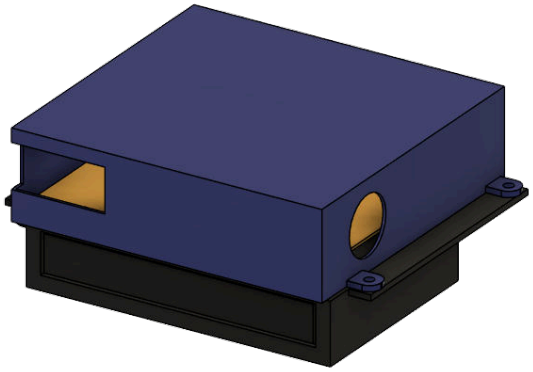


Figure 3: Electronics bus Cover with holes for wiring

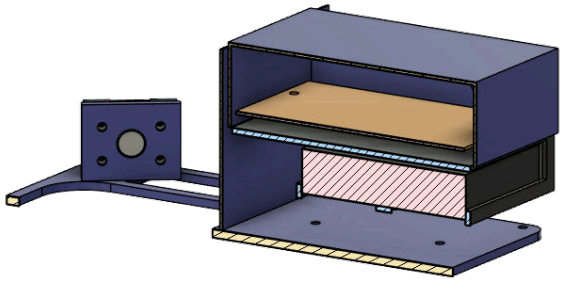


Figure 4: Cross sectional view of the payload including thermal sensor mount



Figure 5: Custom printed battery pack holder

3.2 Software

The mission software is split into two products: the flight software responsible for running all in flight mission operations, and the post-mission data analysis suite. The programming involved in both sections will be performed via Python (with adaptions made for the micropython interface with the onboard computer).

Flight Software:

The flight software runs in one main loop, but is compartmentalised into 3 sections:

1. **pre-loop set-up operations:** SD card initialisation, sensors and operations classes initialization, creation of files (with headers) on the sd card, and setting of flags and initial lists for OBDH.
2. **Data collection segment of main loop:** Data acquisition, data storage, and the triggering check (and associated post-trigger operations if triggered).
3. **Telemetry segment of main loop:** Transmissions are scheduled based on how many iterations of the main loop have been completed, to fulfil timing requirements.

The triggering criteria for releasing the water sample will be checked once per loop, until one of them has been met. At this point, the thermal sensor (MLX90640) will be given an extended number of frames to capture at a higher temporal resolution, in order to maximise scientific data captured.

Figure 6 describes the software flow. Initial set-up tasks are shown in the green section and act before the loop is started. The main loop encompasses the orange and purple sections and will run from after set-up is completed to the moment when the TupperSat is recovered and powered off. More detailed descriptions of the software architecture, as well as the code itself, can be found in the SQRT_Project GitHub repository attached with this data pack and at this link: `https://github.com/Bruce-A-R/SQRT_Project.git`.

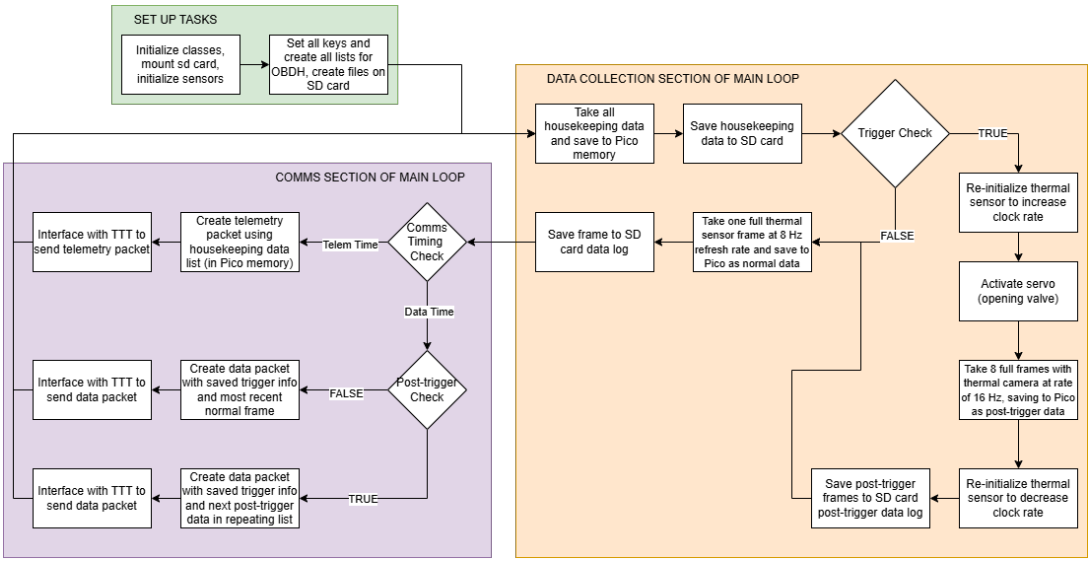


Figure 6: Simplified Software Flowchart

Post-Mission Analysis Software:

The post-mission data analysis suite has been designed with the intention of providing each team member with a uniform dataset with which to complete the final mission report. Processing will involve the removal of any corrupted or missing data and the correlation of machine time to "real" time in the absence of GPS data. This software is included in the GitHub repository linked above.

4 Operator Health & Safety

All SQRT launch and recovery operations shall be carried out in accordance with the approved UCD fieldwork hazards and safe working procedures for TupperSat launch activities. These activities take place outdoors in rural environments and therefore there are some human safety considerations that must be followed by all SQRT launch personnel.

- All operators shall wear suitable outdoor footwear with good grip, as launch and recovery may take place in farmland, forest, bog, hills, or muddy terrain.
- Appropriate weather protection shall be worn, including warm layered clothing, sunscreen, and high-visibility clothing.
- SQRT Personnel assisting with the helium tank and balloon filling should follow launch operator instructions at all times. Latex gloves or other necessary PPE shall be worn when handling the balloon and filling equipment.
- Care should be taken when handling ropes, carabiners, cable ties, structural brackets, and payload mounting hardware to avoid any rope burns, pinching, or minor cuts.
- SQRT Operators shall avoid entering any unsafe terrain, including deep bog, steep slopes, flooded ground, dense forestry, livestock enclosures, or areas near barbed wire fencing unless directed to do so.
- Team members shall bring enough water, food, and any required personal medication due to the long duration of launch and recovery operations and the very real possibility of delayed return travel.

5 Satellite Safety & Handling

This section outlines the procedures that are required to safely store, transport and handle the payload from the delivery stage, the launch stage and the preparation of launch. These outlined procedures are put in place to ensure the payload is optimally protected and flight-ready in the last stages ahead of the mission launch.

5.1 Transport and Storage Advice

- During the TupperSat delivery to the launch operation site, the TupperSat should be carefully transported and stored within a protective container to avoid and prevent any potential mechanical damage, whether to the hardware structure of the payload or the software/electrical elements.
- The payload must be kept in a safe environment and avoid any possible excessive vibration, impact or moisture.
- The payload should remain in 'Power Off' mode for the full duration of transport and should only be powered on during the moment of launch to avoid power drainage of the system or potential fire hazard.
- The payload shall always be handled using both hands and should never be placed under any other loads or beneath additional equipment, this prevents bending or structural damage to the payload.
- The payload should not have any electrical interference in transportation and should be entirely isolated where possible.

5.2 Experimental Safety

- The water container, tubing, and valve mechanism shall remain fully sealed during transport and storage to prevent any possibility of leakage, particularly if the payload is filled prior to departure.
- Care should be taken to ensure that the exposure mechanism is not subjected to unnecessary mechanical stress, vibration, or impact that could disturb the sealed system or accidentally actuate the valve.
- The payload shall be protected from excessive heat, direct sunlight, freezing storage conditions, and high humidity before launch preparation begins.
- If the payload is stored for an extended period prior to launch, the external condition of the water chamber, tubing interfaces, and valve assembly should be rechecked before the payload is accepted for integration.

- If any evidence of water leakage, moisture ingress, condensation, incomplete sealing, or damage to the water containment system is observed, the payload shall not proceed to launch preparation until a full inspection has been completed.
- Any electrical components in proximity to the water system must also be inspected for water exposure or corrosion before power is applied.
- The payload shall remain powered off during transport and storage unless activation is specifically required for an approved pre-launch check. Unauthorised powering of the system is not permitted, as this may result in draining battery power prior to launch and filling SD card memory unnecessarily.
- If the payload has experienced a drop, heavy impact, visible structural damage, or unintended opening of the enclosure, it shall be removed from the launch flow until it has been re-inspected and declared flight-ready by the responsible team member.

THE PAYLOAD SHALL NOT BE LAUNCHED IF ANY OF THE FOLLOWING OCCUR AND ARE NOT ADDRESSED:

- Visible leakage is observed from the water reservoir, tubing, or valve mechanism.
- Mechanical damage to the structure or mounting points observed
- Battery is unsecured or components normally fixed are loose.
- Evidence is seen that the valve has actuated (water exposed).

6 Operations Procedures

The procedures for operation of the satellite may be found in the Appendix below.

This includes the procedures for:

- **A. Final Assembly and Handling Documentation.**
  This section covers all assembly considerations and steps required prior to the transportation and delivery of the SQRT TupperSat to the launch team, and details relating the handling and transportation of the satellite to the launch site.
- **B. Launch Day Tools & Equipment.**
  This section outlines all tools and equipment required at the launch site for routine final checks and for any troubleshooting in case of any problems.
- **C. Launch Documentation.**
  This section outlines all actions to be followed at the launch site to prepare and launch the satellite, all final flight tests to be performed and a detailed list of go/no–go conditions relating to the launch of the SQRT TupperSat.
- **D. Recovery Documentation**
  This section describes all actions and steps to be followed to deactivate the satellite upon recovery.
- **E. Flight Operations Documentation**
  This section lists the actions that must be performed during flight.

7 Appendix

A Final Assembly & Handling Documentation

This section covers all assembly considerations and steps for connecting components to the perfboard during assembly in addition to the structural and hardware component assembly required prior to the final delivery of the SQRT payload to the launch site.

Information for Connecting Sensors to Perfboard: Steps 20 through 27 reference connection points labelled in Figure 10, which shows the perfboard as mounted on the battery pack without the connections to outside components plugged in. On Figure 10, the red L shape marked 1 represents the rectangular hole that slightly wraps around a corner and must be situated at the corner next to connection F. The light blue rectangle marked 2 represents the circular hole in the blue covering that must be situated in front of connection E. In the following procedure these holes will be called Hole 1 and Hole 2. Connections are referred to by the letters shown on Figure 10 (i.e. Connection A for the connection labelled A). Figure 12 shows connections as they are made. The left image shows connections through Hole 2, and the picture on the right shows all connections connected (through both Hole 1 and Hole 2).

A full video tutorial of the payload assembly may be found at the following link:
`https://youtu.be/cfUkccYLhPU?si=hDir8J9pxijI3R3K`

A.1 Full Procedure and Instructions for Assembly

Table 4: SQRT Assembly Procedure

Task	**Time Completed**	**PA Signature**
Lay out all components of payload clearly and ensure the necessary tools are to hand and are accounted for as per the full component image above (See Figure 1 for components).		
Unscrew the blue cap from the end of the syringe water container then fill the container with hot water such that it is full except for <0.25 ml of an air bubble.		
Screw the blue threaded cap onto the threaded connection end of the syringe until tight.		
Attach the TTT antenna to the bottom of the Tupperware box using the four allocated drilled holes at its base.		
Ensure the communications wire of the antenna is fed through the allocated hole in the middle of the box bed and thereafter leave cable aside for now.		
Tightly secure using the four smaller threaded rods and 3mm nuts, this will need to be completed both at the base of the antenna and at the blue printed base on the bed of the Tupperware box.		
Thereafter, place the four larger 131mm threaded rods minto the four corner holes of the system and secure them from underneath using the 3mm nuts.		
Once rods are in place, insert the separation wall by jimmying the two allocated screw holes onto the two inner thread walls.		
Attach the MLX90640 thermal sensor (in its case packaging) onto the two holes in the blue mount within the low pressure chamber (See Figure 4) by pushing the two 2.5mm diameter bolts attached to the sensor mount through the holes in the blue mount and out through the bottom of the tupperware, then fixing them in place with 2.5mm nuts tightly.		

Thread the white cable coming up from the antenna through the hole in the bottom of the reflective thermal insulation, then place the thermal insulation into the payload such that it sits directly over the antenna connection within the tupperware, such that it fits inside of the payload as seen in Figure 7.		
Thread four blue-green spacers (the second longest set of spacers) onto the four threaded rods (one for each rod).		
Carefully place the tubing of the water container, with a securely sealed water sample inside its cylinder, into the valve lock of the servo motor.		
Ensure the servo-motor valve is in the locked position before placing it carefully into the TupperSat atop the blue base and layer of insulation. The tubing and part of the printed valve structure should feed through the small cut hole in the separating wall. If done correctly, it should be secure and unable to move from its position by the two centre threaded rods. This placement is shown in Figure 8.		
Activate and position the heated pad alongside the syringe, ensuring that it is positioned such that the battery housing frame can sit flush on the spacers.		
Before continuing, check that the water container cap and pulling string are positioned toward the back of the TupperSat payload, pressed to the inside of the reflective insulation.		
Activate and insert a small heating pad inside the base layer along the syringe and place the internal temperature sensor nearby toward the syringe-side wall to avoid interaction with the servo-motor. This placement is shown in Figure 9.		
Slot the power pack carefully inside the printed power pack holder slot which is directly underneath the perfboard with the electronics, such that the power button is facing outwards from the pack (accessible while the batter is inside) and the USB ports are also accessible.		
Connect the USB-A side of the black USBA - micro USB power cable to the battery pack.		
Screw the perfboard onto the black battery pack via the holes in the perfboard into the mounted cylinders on top of the battery pack (opposite face from the battery), aligning the perfboard such that the Pico is aligned lengthwise with the side of the battery pack with USB ports. This alignment is shown in Figure 10, where the USBA - micro USB connection can be seen at the top of the image.		
Connect the external ds18b20 temperature sensor (the sensor with the silver cylinder at one end) to the female headers on the perfboard at connection A. The thermal sensor has three male pin connections at the end. Thread the end of the temperature sensor with the pin connections through Hole 1 in the blue cover such that the sensing end of the temperature sensor is outside the cover. Connect the temperature sensor pins to the three female connections in Connection A such that the black cable points towards the top of the image in Figure 10. Yellow tape covering the pins should be above the location of Connection A on the image in this orientation.		

Connect the internal temperature sensor (ds18b20 with tiny black half-cilynder at one end) to the female headers such that the yellow wire coming from the sensor aligns with the yellow wire on the perfboard and the black wire on the sensor aligns with the black wire on the perfboard. Thread the sensor head (black half-cylinder) through Hole 1 in the blue cover out from inside the perfboard such that the sensor head is now outside of the blue cover while the other end of the wires are still connected to the perfboard.		
Connect the thermal sensor wire bundle to Connection B on the perfboard by threading the connecting end of the wire bundle (the loose end with a black piece) through Hole 2 then connecting the wire connection to Connection B such that the white wire from the thermal sensor is lined up with the white wire leading from Connection B (On Figure 10, this wire can be seen directly under the circle with B in it).		
Connect the female connectors on the servo motor wires to Connection C (shown in Figure 10) such that the brown wire from the sensor connects to the black wire on the perfboard. This connection orientation can be seen in the left image of Figure 12.		
Connect the GPS antenna to the module that sits connected to the perfboard via Connection D by first threading the cable from the GPS antenna through Hole 2 from the outside such that the right-angled gold connector will be inside the cover and the antenna will be outside when the cover is fully attached. Then, screw the right-angled gold-coloured connection to Connection D until hand tight, such that the black wire directly exits Hole 2.		
Connect the TTT antenna wire to the TTT module on the perfboard via Connection E by doing the following: A white wire is connected to the antenna via the bottom of the Tuppersat, with a gold connection point on the loose end. Thread this wire up and through the outside of the blue cover through Hole 2. Attach the connector to Connection E by threading it on until hand tight.		
Orient the blue cover in position, such that the cables running through Hole 1 and Hole 2 are still out of the holes and the three small holes on the bottom of the cover align with three of the four holes in the black base the perfboard is attached to. The Pico's power port and Hole 1 should be positioned in the corner of the TupperSat diagonally opposite the thermal sensor, such that the port is facing the external wall of the structure.		
Connect the micro usb plug coming from the battery pack to the Pico onboard computer at Connection F by plugging the micro-USB connection to the micro-USB port in the Pico through Hole 1. This micro-USB connector is on the black with one USB A male end already plugged into the battery pack, and one micro USB male end that is at a right angle from the cable.		
Press the button on the bottom of the battery pack once to turn on the power supply.		
Confirm that power is reaching the Pico by observing that there is a red light coming from the perfboard visible through Holes 1 or 2.		

Confirm that power is reaching the Pico by observing that there is a red light coming from the perfboard visible through Holes 1 or 2.		
Wait up to 60 seconds with a receiving ground station and confirm that data packets have been transmitted to confirm correct power-on.		
Thread the four threaded rods through the four holes in the battery pack, and the 3 holes of the oriented blue cover (oriented such that each of the three holes is directly on top of one of the holes in the black cover and the blue cover is oriented so the GPS and TTT antena cables directly leave Hole 2), such that the battery pack sits on top of the four long blue spacers.		
Thread the small blue washer onto the rod that went through the battery pack but had no blue cover hole to go through.		
Thread one of each of the remaining blue 40mm spacers onto each of the four threaded rods.		
Thread one of each of the remaining blue 40mm spacers onto each of the four threaded rods.		
Insert any remaining insulation material necessary such as the rectangular and square wall insulators.		
Feed the external temperature sensor through its allocated slot on the red lid such that the entire silver end sticks out of the red lid, and tuck remaining black wire between solver insulation and the inner wall of the plastic tupperware container such that it will not get in the way of the red lid flaps.		
Carefully place the lid atop the GPS antenna which should sit on the electronics lid and sit snug in the hole on the red lid (overall TupperSat lid).		
Fit the red lid to the top of the structure, ensuring that each of the 4 threaded rods is fed through their respective holes. The position of the holes with threaded rods is shown in Figures 15 and 16.		
Close the red lid flaps securely around the edges of the Tupperware box,ensure the lid corner holes are secured by the corner threaded rods of the payload.		
Tighten the four remaining 3mm nuts around the threaded rods protruding from the red lid until they are flush with the red lid and tight.		
Secure the four eye-nuts to the top of the payload by rotating them clockwise onto the corner threaded rods until tight.		
Tie the rope used to connect the Tuppersat to the balloon through each eye-nut to form an 'X' shape over the top of the payload, then tie a loop used to connect with the balloon.		

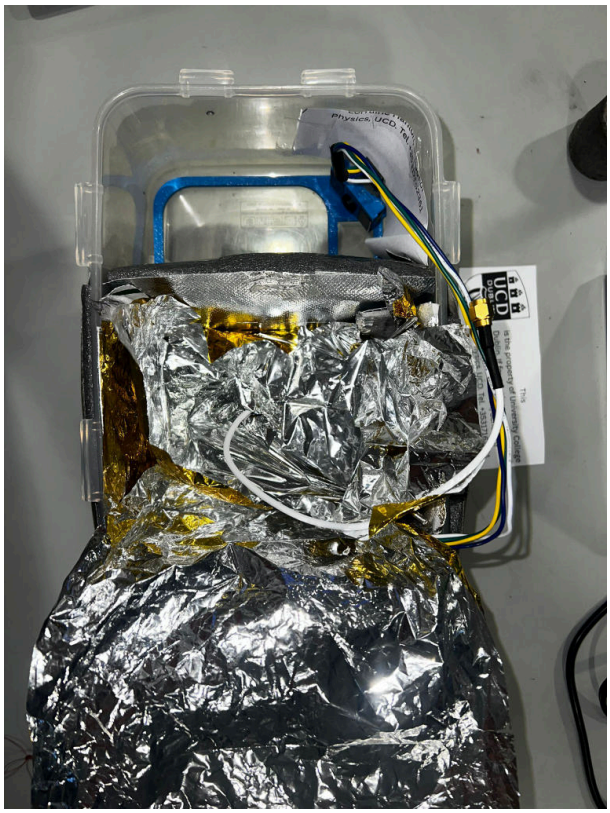


Figure 7: Assembly of TupperSat before inclusion of water container and servo-motor, electronics bus, sensors/electric components, and heat pads

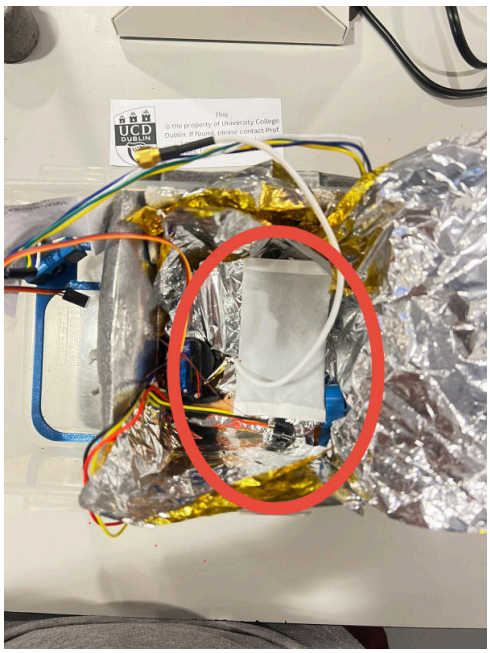


Figure 9: Assembly view with heat pad and temperature sensor in place near the syringe

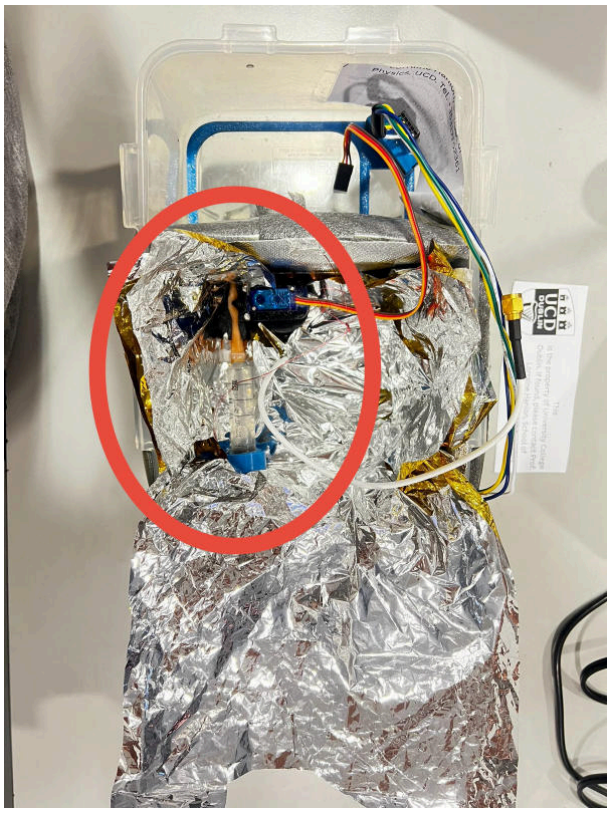


Figure 8: Assembly with servo-motor and water container in place



Figure 11: The electronics perfboard screwed atop the battery holder

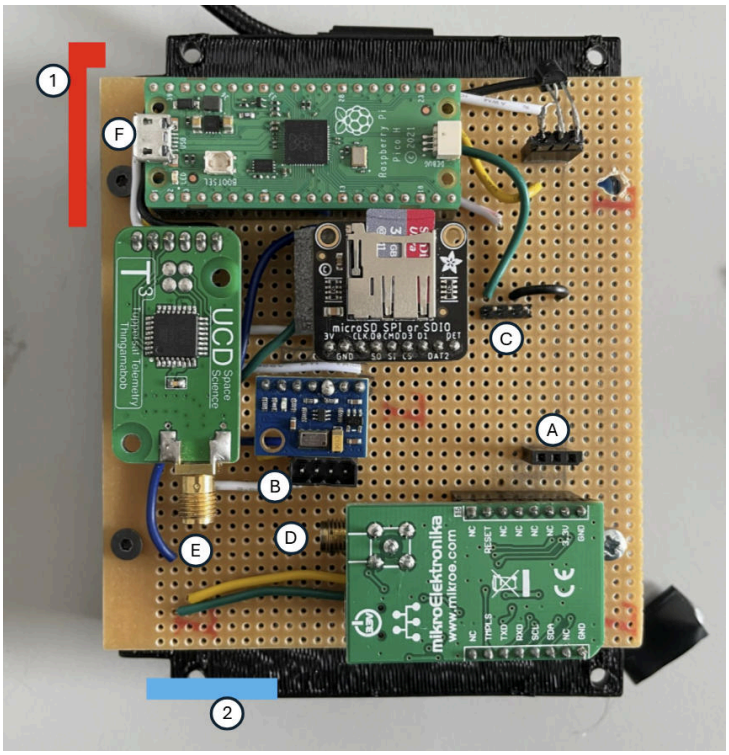


Figure 10: Labelled Perfboard Connections

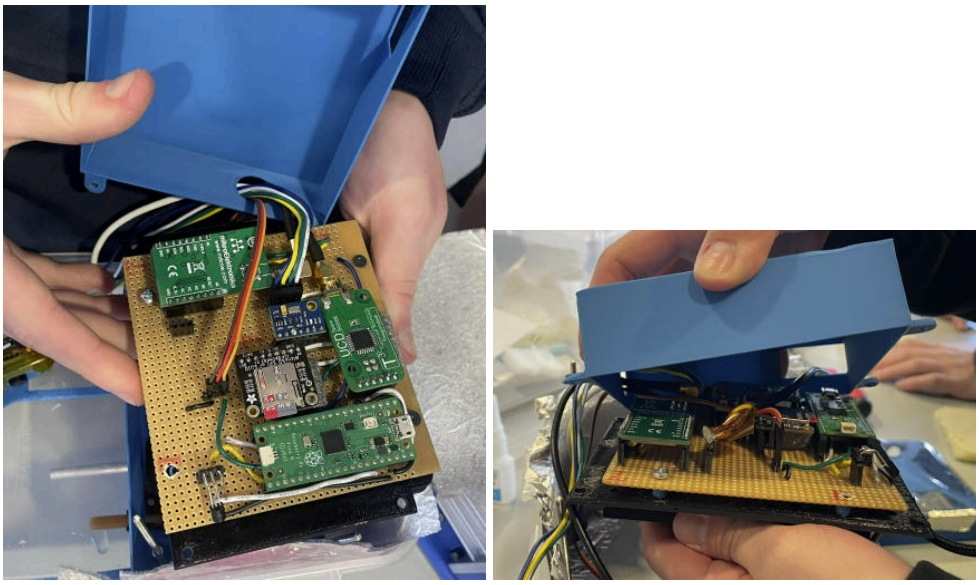


Figure 12: Perfboard With Connections

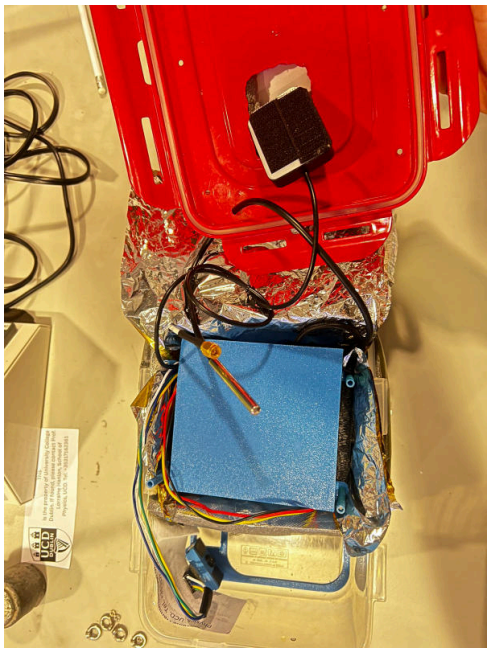


Figure 13: Assembly with electronics cap and spacers



Figure 14: GPS seated location with external temperature sensor



Figure 15: Location of 3mm nuts and threaded bars for top view of payload



Figure 16: Eye-nuts on payload threaded rods to be used for balloon support

B Launch Day Tools & Equipment

The equipment listed in Table 5 should be brought along to the launch site to facilitate pre-flight operations and, if necessary, trouble-shooting.

Table 5: Launch Day Equipment

Equipment	Amount	Justification
Water	>= 0.5 L	Use to fill/refill water container as necessary.
Charged Laptops	2	Needed as ground stations.
Charged Phones (belonging to team members)	5	For use communicating during launch and flight operations.
USB A - micro USB connection cable	2	Bring extra cables in case of breakage, looseness, or loss of equipment.
Fully charged powerbanks	2	Shall be taken to ensure all equipment can be charged.
Philips head screwdriver	1	Used to adjust screws if necessary.
Duct or electrical tape	1 roll	For use securing flaps of insulation, reenforcing tuppersat lid connection, or anything else that may arise.
Silicon tubing	80mm	Bring in case tube must be replaced.
Printed launch documents packet, including pre-flight checklist	1	For pre-flight checklist and other documentation needs.
Pens/pencils	10	To write things as needed.

C Launch Documentation

This section outlines all of the actions that should be followed at the launch site to prepare and launch the SQRT TupperSat, the final flight tests that need to be carried out, and the detailed GO/NO-GO conditions that decide whether the payload is safe and ready to fly.

C.1 Launch Site Preparation

1. Once we arrive at the launch site, the team should ensure that the SQRT payload remains powered off while the balloon, ground station, and recovery setup are being prepared.
2. Before doing anything else, the full payload should be given a careful visual check to ensure nothing was damaged during transportation, especially within the Tupperware structure, on the TTT antenna, at the rope attachment eye-nut points, and the external temperature sensor. Preparation can proceed once team members have confirmed that pieces are not missing or damaged.
3. Once the external and internal structure is confirmed to be complete and undamaged, the water reservoir should be refilled with the water sample. The syringe, tubing, and valve area should then be checked carefully for any sign of leakage.
4. The servo pinch valve should then be checked to make sure it is fully closed and still in the safe pre-trigger position. The vent opening to the low-pressure chamber should also be checked to make sure it has not become blocked during transport or refilling.
5. After the physical checks are complete, the power pack can now be connected and the electronics payload may be powered on.
6. At this stage the SQRT team should confirm that the Raspberry Pi Pico boots up correctly and the telemetry link is working by successfully receiving downlinked data and confirming downlinked values are as expected.
7. The payload can then be attached to the balloon using the approved mounting point and carabineer setup.
8. Before release, one final full visual check shall be done to ensure external connections are secure before the balloon is launched.

C.2 Final Flight Checks and GO-NO GO Conditions

The final pre-flight checks listed below should be completed immediately before the balloon launch, after the Tuppersat has been powered on and is completely assembled. Table 6 describes five GO and NO GO conditions to check before flight. All five conditions must be in the GO state for the satellite to be launched.

- Confirm that weather is still good for launch (see Table 6)
- Confirm both a data packet and a telemetry packet have been successfully downlinked within 30 seconds of powering on the Tuppersat with a receiving ground station ready.
- Confirm that data packets downlinked contain appropriate housekeeping and science data that indicated functionality of sensors and code (Expected GPS location, temperature sensor temperatures, pressure sensor pressure, trigger check/condition values).

The final decision to launch the SQRT TupperSat shall be made only once all final pre-flight checks have been completed and the payload is confirmed to be safe, functional, and fully flight-ready.

The following **GO & NO-GO** conditions must be considered in advance of release.

Table 6: Launch Day Equipment

Condition	GO	NO GO
1	Weather conditions are sunny and suited for high altitude balloon flight (i.e. minimal wind, cloud cover and rain)	Weather condition are unsuitable for flight (i.e. rain or windy weather that would damage the balloon).
2	There is no visible structural damage to payload components	There is evidence of structural damage (i.e. missing components, pieces missing from components, cracks).
3	Water has not leaked from the container and/or valve (either by broken part or unclosed valve)	There is loose water in the payload (can be visible in the low pressure chamber through the clear tupperware plastic).
4	The battery pack is fully charged (3 dots light up when button pressed) or at a charge confirmed to last 6 hours with full operations	The battery pack is not fully charged (less than three dots light up when button pressed).
5	Data and Telemetry packets are successfully communicated on the ground between powered-on satellite and ground station	Communications are not received, packets are unknown rather than Data and Telemetry.

D Recovery Documentation

If the satellite is recovered, special care has to be taken of the payload. Consider that this TupperSat contains around 4ml of liquid water in a sealed container at launch, but if the experiment is successful, it will be spread all over the tupperware during descent. This means that the water will eventually condensate and become liquid, moving around inside the plastic frame. The design was done considering this, but once it's recovered, the following steps should be taken to reduce the risk of water affecting the onboard SD card: the Tuppersat should be held in such a position that the water is the furthest possible from the electronic bus and thermal sensor. Hold the Tuppersat upright such that the red cover is pointing up, and hold from the corner circled in Figure 17.

To deactivate the Tuppersat, the lid has to be taken out after unscrewing the 4 eyenuts and their screws. Once opened, the thermal insulation has to be taken away, by pulling carefully from the top part. Figure 18 shows the Tuppersat without the lid. The left picture shows the thermal cover on, and the right one shows the thermal cover off after it has been peeled up and away.

Once removed, a black cable with a rope-like appearance can be seen in a right-angle connection (red circled in the right picture of Figure 18). That wire connects the electronics to the battery. The Tuppersat is deactivated by disconnecting that wire, which can be gently pulled outwards from the blue casing.

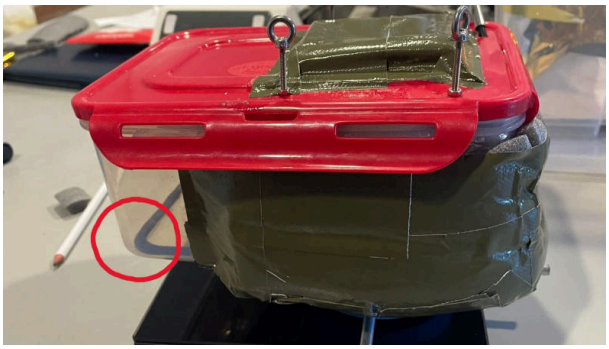


Figure 17: How to hold Tuppersatsat once recovered

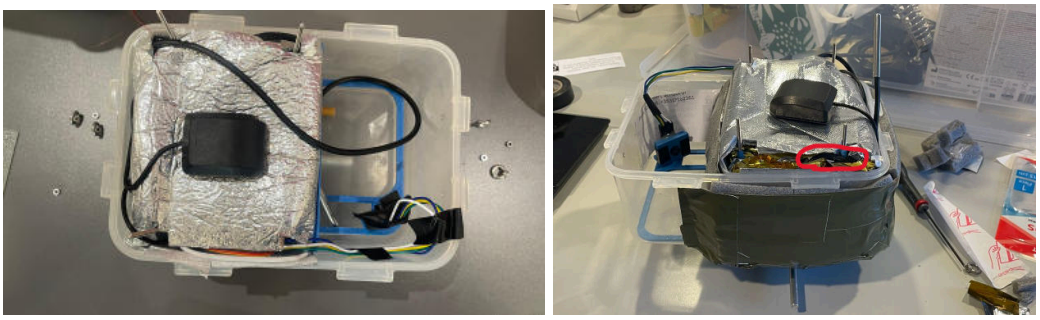


Figure 18: Thermal isolation removal

Disassemble the Tuppersat further by reversing the assembly steps. Remove the SD card from the SD card (red and gray) from the SD card holder (located at the centre of the perfboard) to collect the data saved onboard.

E Flight Operations Documentation

This section describes all actions that must be performed during flight:

- Use ground station to listen for transmissions from the TupperSat (and other TupperSats in flight) and record downlinked data.
- Follow the track of the balloon using the predicted flightpath and downlinked position data to position team members close to the expected drop zone for recovery.

F Contact Personnel

In the event that the SQRT mission team must be contacted for any reason. Please email the following primary and secondary contact points:

1. **Elyana Abdallah**
2. **Bruce Ritter**